Exercise 1

We first load a dataset and examine its dimensions.

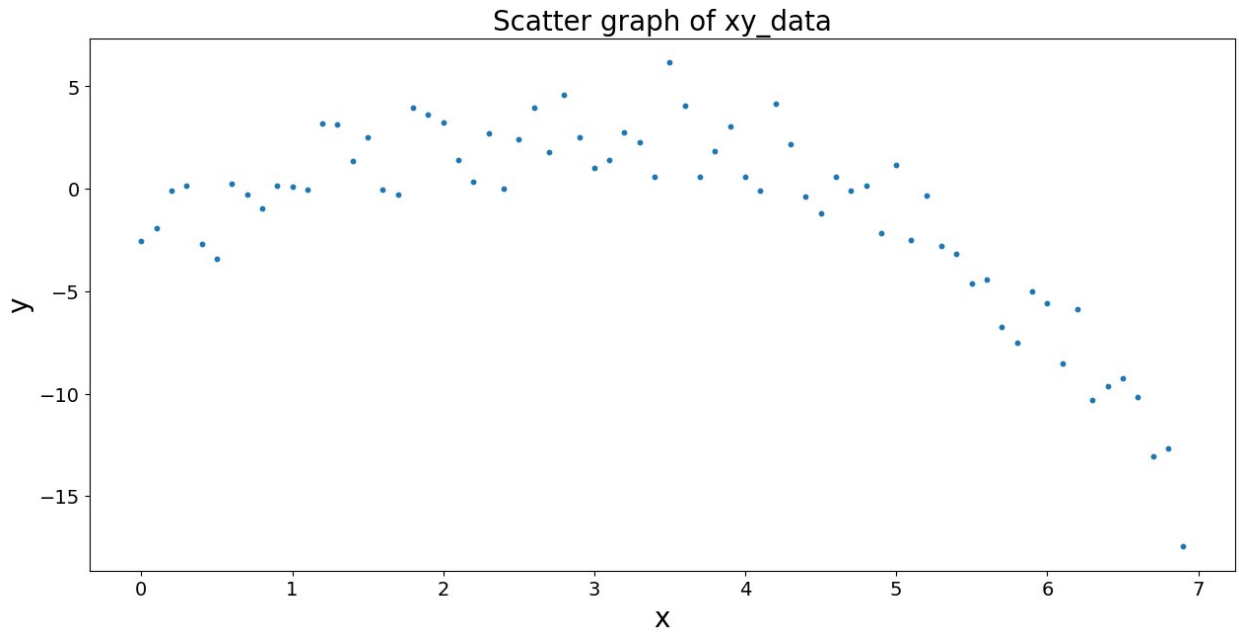
```
# If you are running this on Google Colab, uncomment and run the
following lines; otherwise ignore this cell
# from google.colab import drive
# drive.mount('/content/drive')

import math
import numpy as np

xy_data = np.load('Ex1_polyreg_data.npy')
# If running on Google Colab change path to
'/content/drive/MyDrive/IB-Data-Science/Exercises/Ex1_polyreg_data.npy
'

np.shape(xy_data)
```

```
(70, 2)
```

The matrix `xy_data` contains 70 rows, each a data point of the form (x_i, y_i) for $i=1,\ldots,70$.

1a) Plot the data in a scatterplot.

```
import matplotlib.pyplot as plt

x_data = xy_data[:,0]      # First column of array
y_data = xy_data[:,1]  # Second column of array



plt.rcParams['figure.figsize'] = [15, 7]
plt.rcParams['axes.titlesize'] = 20
plt.rcParams['axes.labelsize'] = 20
plt.rcParams['xtick.labelsize'] = 14
plt.rcParams['ytick.labelsize'] = 14

plt.scatter(x_data, y_data, s=10)   # s can be used to adjust the size
of the dots
plt.xlabel('x')
plt.ylabel('y')
plt.title('Scatter graph of xy_data')

plt.show()
```



1b) Write a function `polyreg` to fit a polynomial of a given order to a dataset.

The inputs to the function are a data matrix of dimension $N \times 2$, and $k \geq 0$, the order of the polynomial. The function should compute the coefficients of the polynomial $\beta_0 + \beta_1 x + \ldots + \beta_k x^k$ via least squares regression, and should return the coefficient vector, the fit, and the vector of residuals.

If specified the degree k is greater than or equal to N, then the function must fit an order $(N-1)$ polynomial and set the remaining coefficients to zero.

NOTE: You are *not* allowed to use the built-in function `np.polyfit`.

```
def polyreg(data_matrix, k):

    N = np.shape(data_matrix)[0]

    if k<N:
        order=k
    else:
        order=N-1

    x_data = data_matrix[:,0]
    y_data = data_matrix[:,1]


    X = np.column_stack([x_data**i for i in range (0,order+1)])
    XT= X.T

    betas = np.linalg.inv(XT.dot(X)).dot(XT.dot(y_data))
```

```
    fit = X.dot(betas)
    resid = y_data - fit


    #appending additional zeros
    betas=np.append(betas,[0 for i in range(order,k)])
    # The function should return the the coefficient vector beta, the
fit, and the vector of residuals
    return(betas,fit,resid)
```

Use the tests below to check the outputs of the function you have written:

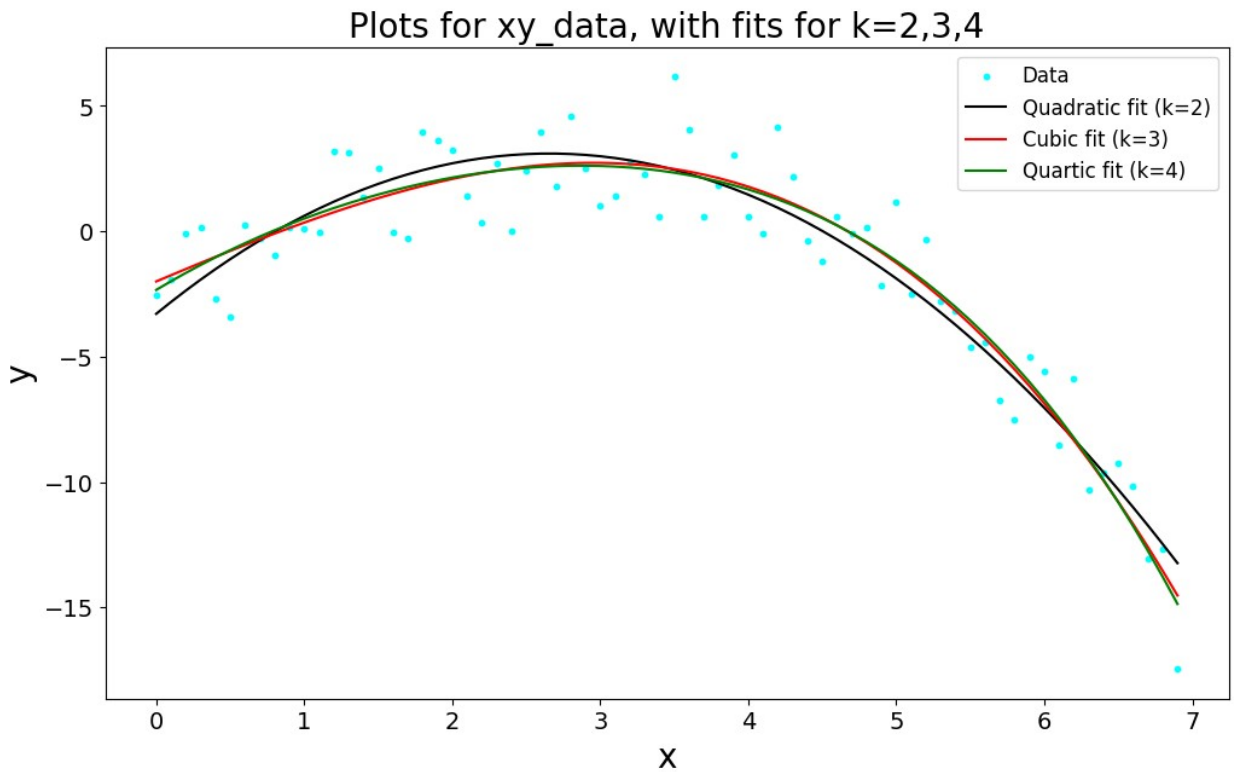
```
# Some tests to make sure your function is working correctly

xcol = np.arange(-1, 1.05, 0.1)
ycol = 2 - 7*xcol + 3*(xcol**2)  # We are generating data accoridng to
y = 2 - 7x + 3x^2
test_matrix = np.transpose(np.vstack((xcol,ycol)))
test_matrix.shape

beta_test = polyreg(test_matrix, k=2)[0]
assert((np.round(beta_test[0], 3) == 2) and (np.round(beta_test[1], 3)
== -7) and (np.round(beta_test[2], 3) == 3))
# We want to check that using the function with k=2 recovers the
coefficients exactly

# Now check the zeroth order fit, i.e., the function gives the correct
output with k=0
beta_test = polyreg(test_matrix, k=0)[0]
res_test = polyreg(test_matrix, k=0)[2] #the last output of the
function gives the vector of residuals

assert(np.round(beta_test, 3) == 3.1)
assert(np.round(np.linalg.norm(res_test), 3) == 19.937)
```

1c) Use `polyreg` to fit polynomial models for the data in `xy_data` for $k=2,3,4$:

- Plot the fits for the three cases on the same plot together with the scatterplot of the data. The plots should be labelled and a legend included.
- Compute and print the SSE and R^2 coefficient for each of the three cases.
- Which of the three models you would choose? Briefly justify your choice.

```
fit_0 = np.full_like(y_data, np.mean(y_data))
betas_2,fit_2,resid_2 = polyreg(xy_data, k=2)   # Computing the fit of
the quadratic model
betas_3,fit_3,resid_3 = polyreg(xy_data, k=3)   # Computing the fit of
the cubic model
```

```
betas_4,fit_4,resid_4 = polyreg(xy_data, k=4)   # Computing the fit of
the quartic model

plt.rcParams['figure.figsize'] = [12, 7]

plt.scatter(x_data, y_data, s=10, color = 'cyan', label = 'Data')
plt.plot(x_data, fit_2, color = 'black', label = 'Quadratic fit
(k=2)')
plt.plot(x_data, fit_3, color = 'red', label = 'Cubic fit (k=3)')
plt.plot(x_data, fit_4, color = 'green', label = 'Quartic fit (k=4)')

plt.xlabel('x')
plt.ylabel('y')
plt.title('Plots for xy_data, with fits for k=2,3,4')
plt.legend(fontsize = 'large')

plt.show()


#We now compute the sum of squared errors
SSE_0 = np.linalg.norm(y_data - fit_0)**2
SSE_2 = np.linalg.norm(y_data - fit_2)**2
SSE_3 = np.linalg.norm(y_data - fit_3)**2
SSE_4 = np.linalg.norm(y_data - fit_4)**2

print('SSE_0 = ', np.round(SSE_0, 2))
print('SSE_2 = ', np.round(SSE_2, 2), ' \t R^2_2 = ', np.round(1-
SSE_2/SSE_0, decimals = 4))
print('SSE_3 = ', np.round(SSE_3, 2), ' \t R^2_3 = ', np.round(1-
SSE_3/SSE_0, decimals = 4))
print('SSE_4 = ', np.round(SSE_4, 2), ' \t R^2_4 = ', np.round(1-
SSE_4/SSE_0, decimals = 4))
```




```
SSE_0 =  1532.27
SSE_2 =  172.18        R^2_2 =  0.8876
SSE_3 =  152.41        R^2_3 =  0.9005
SSE_4 =  151.23        R^2_4 =  0.9013
```

State which model you choose and briefly justify your choice.

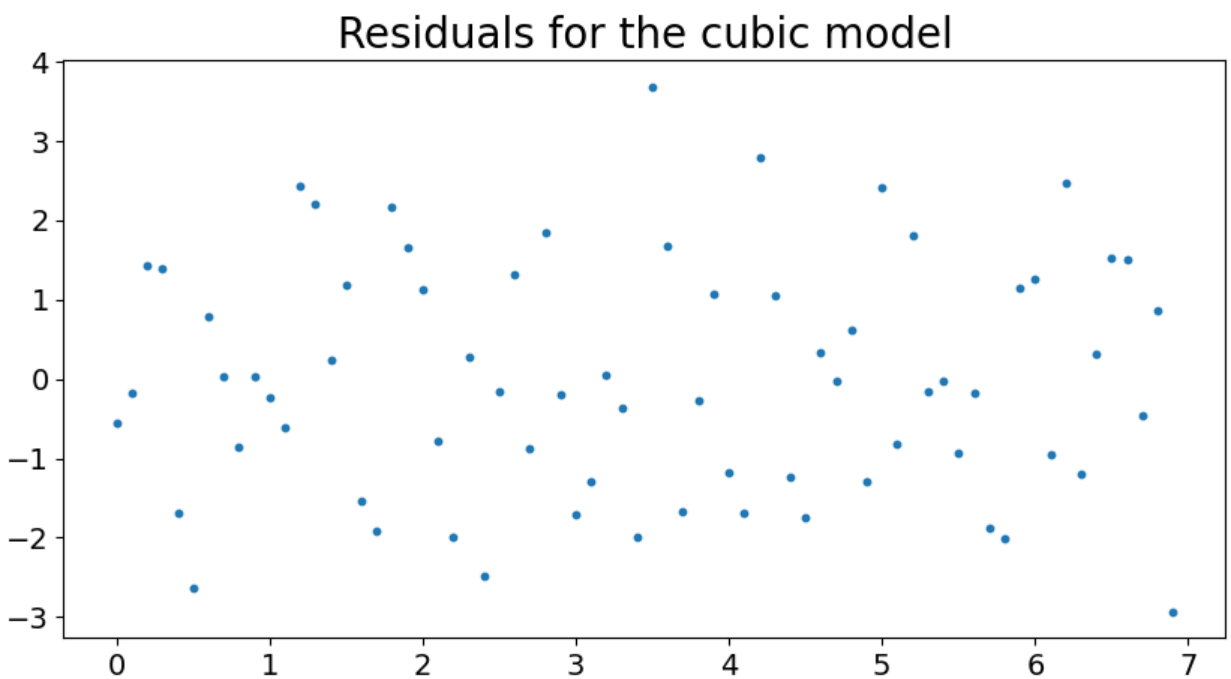
The best model to use would probably be the cubic model, as it offers a noticable improvement in accuracy compared to the quadratic model, and is similar in goodness of fit to the quartic model - while being simpler. The simplest model with the best accuracy is most valuable for predicting future data.

1d) For the model you have chosen in the previous part (either $k=2/3/4$ ¿:

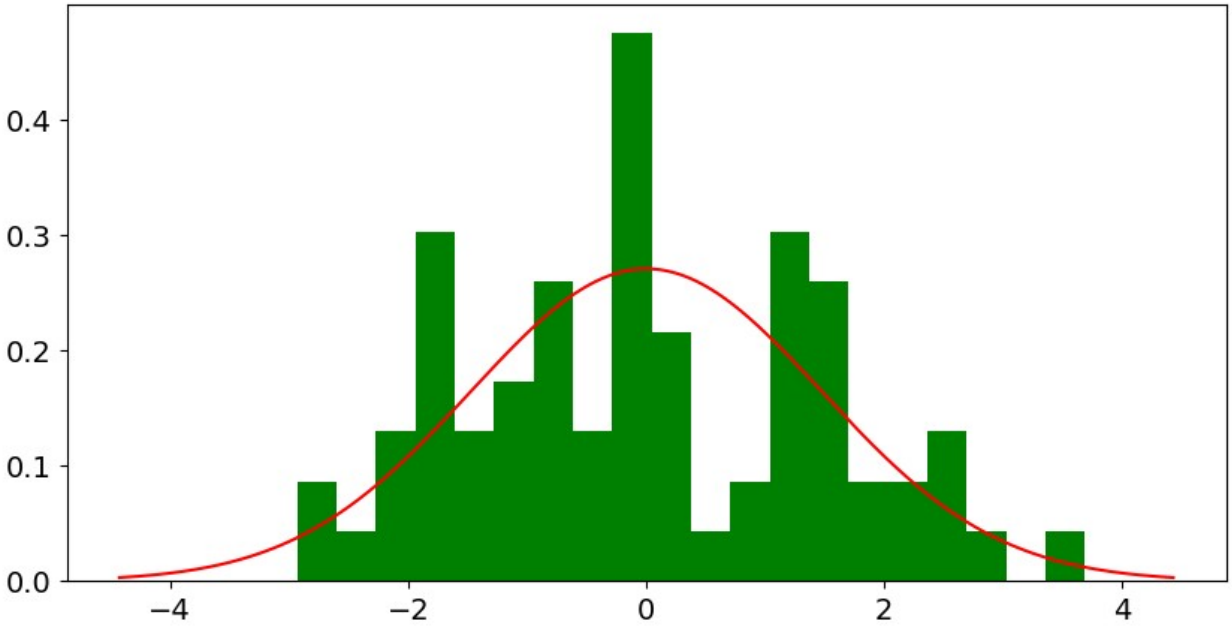
- Plot the residuals in a scatter plot.
- Plot a histogram of the residuals along with a Gaussian pdf with zero mean and the same standard deviation as the residuals.

```
from scipy.stats import norm

plt.rcParams['figure.figsize'] = [10, 5]
plt.scatter(x_data, resid_3, s=10)
plt.title('Residuals for the cubic model')

plt.show()

print('Mean of residuals for cubic model = ',
```

```
np.round(np.mean(resid_3), 3),
      'Variance of residuals = ', np.round(np.var(resid_3),3))

# Plot normed histogram of the residuals
n, bins, patches = plt.hist(resid_3, bins=20, density=True,
facecolor='green');

# Plot Gaussian pdf with same mean and variance as the residuals
res_quad_stdev = np.std(resid_3)  #standard deviation of residuals
xvals = np.linspace(-3*res_quad_stdev,3*res_quad_stdev,1000)
plt.plot(xvals, norm.pdf(xvals, loc=0, scale=res_quad_stdev), 'r')
plt.show()
```




```
Mean of residuals for cubic model =  0.0 Variance of residuals =
2.177
```


0.4
0.3
0.2
0.1
0.0
−4
−2
0
2
4